

Question 1

a) $9280h = 1001\ 0010\ 1000\ 0000_2$
 $903Ah = 1001\ 0000\ 0011\ 1010_2$

$$\begin{array}{r}
 \overset{''}{1001}\ 0010\ 1000\ 0000 \quad (9280h) \\
 + \quad 1001\ 0000\ 0011\ 1010 \quad (903Ah) \\
 \hline
 1\ 0010\ 0010\ 1011\ 1010 \quad (22BAh)
 \end{array}$$

(CF)

ZF	SF	CF	OF	AF	PF
0	0	1	1	0	1

b)

$8743h = \underbrace{1000}_{4\text{-bit opcode}} \underbrace{0111\ 0100\ 0011}_{12\text{-bit operand}}$

IP/PC	MAR	MBR	IR	BX	COA	Micro-operation
409				0003	0005	IP/PC = 409
40A	40A			0003	0005	MAR ← PC
40B	40A	8743		0003	0005	MBR ← M[MAR], PC ← PC + 1
40B	40A	8743	8743	0003	0005	IR ← MBR
40B	743	8743	8743	0003	0005	MAR ← IR[11:0]
40B	743	0005	8743	0003	0005	MBR ← M[MAR]
40B	743	0008	8743	0003	0005	MBR ← MBR + BX
40B	743	0008	8743	0003	0008	M[MAR] ← MBR

Question 2

a. i) $t_n = 57 + 25 + 45 + 35 + 45 + 40 + 5 = 252 \text{ ns}$

$$T_n = 252 \times 1500$$

$$= 378,000 \text{ ns}$$

ii) $t_p = 57 + 5 = 62 \text{ ns}$

$$T_p = t_p (K + (n-1))$$

$$T_p = 62 \times (6 + (1500-1))$$

$$= 93,310 \text{ ns}$$

iii) $S_{\max} = K$

$$S_{\max} = 6$$

iv) $S = \frac{T_n}{T_p} = \frac{378,000}{93,310} = 4.05$

b. i)

	T_1	T_2	T_3	T_4	T_5	T_6	T_7	T_8	T_9	T_{10}	T_{11}
T_1	FI	DI	FO	EI	SO						
T_2		FI	DI	FO	EI	SO					
T_3			FI	DI	FO	EI	SO				
T_4				FI	DI	X	X	X			
T_5						FI	DI	FO	EI	EI	

ii)

	T_1	T_2	T_3	T_4	T_5	T_6	T_7	T_8	T_9	T_{10}	T_{11}
T_1	FI	DI	FO	EI	SO						
T_2		FI	DI	FO	EI	SO					
T_3			FI	DI	FO	EI	SO				
NOP				FI	DI	FO	EI	SO			
NOP					FI	DI	FO	EI	SO		
NOP						FI	DI	FO	EI	SO	
NOP							FI	DI	FO	EI	SO
T_5							FI	DI	FO	EI	SO